Chapter 8
Moana Matters: Culture and Deep Sea Mining in Oceania

Elise Huffer and Renee Hill-Lewenilovo

Abstract Cultural connectivity has been a marker of Oceania since time immemorial. Islands were fished out of the water by ancestral demi-gods; sea routes were mapped and shared with kindred animals; currents, winds, and stars were felt, breathed in, and drawn in sand and in minds. Sharks, dolphins, and turtles are still called today across different islands Place names are shared across vast expanses and ancient sacred complexes unite communities from different corners of Oceania. Deep sea mining (DSM) threatens this long standing connectivity, compounding the already serious impacts of climate change which are damaging the ocean and islands.

Keywords Deep Sea Mining (DSM) · Oceania · Cultural connectivity · Traditional ecological knowledge · Spiritual turmoil

> Oceania is vast, Oceania is expanding, Oceania is hospitable and generous, Oceania is humanity rising from the depths of brine and regions of fire deeper still, Oceania is us. We are the sea, we are the ocean…
>
> —Epeli Hau'ofa, We Are the Ocean: Selected Works

E. Huffer (✉)
University of the South Pacific, Suva, Fiji

R. Hill-Lewenilovo
Ocean Voices Fellow (2024–2025), Suva, Fiji

© The Author(s) 2026

C. Jarvis (ed.), *Threats to Our Ocean Heritage: Deep Sea Mining*,
SpringerBriefs in Archaeology, https://doi.org/10.1007/978-3-031-98238-5_8

8.1 Introduction

Epeli Hau'ofa's often quoted 'we are the sea, we are the ocean', affirms the depth of connection of the people of Oceania[1] with the ocean. Cultural connectivity has been a marker of Oceania since time immemorial. Islands were fished out of the water by ancestral demi-gods; sea routes were mapped and shared with kindred animals; currents, winds, and stars were felt, breathed in, and drawn in sand and in minds. Sharks, dolphins, and turtles are still called[2] today across different islands (Torrente et al., 2018). Place names are shared across vast expanses and ancient sacred complexes unite communities from different corners of Oceania. Deep sea mining (DSM) threatens this long-standing connectivity, compounding the already serious impacts of climate change which are damaging the ocean and islands. As the region and global community collectively confront DSM, to understand the cultural perspective we turn first to Oceanian scholars, activists, artists, navigators, cultural experts, theologians, and practitioners for guidance. They are telling stories that explain and reinforce traditional bonds to the moana. They provide accounts of kinship and retell ocean histories, genealogies, and contemporary navigational feats. There are many to name and to listen to.

A few examples include Tongan academic Tēvita 'O Ka'ili, whose work on Tongan genealogy illustrates the intimate and enduring relationships with ocean ancestors and the relevance of these to ecological stewardship (Ka'ili, 2024; *Lala's LALINOA*, 2024); Māori scholar Dan Hikuroa, whose writing on mātauranga Māori and kaitiatikanga demonstrates how the principles and values that underpin them, are key to understanding the depth of caring for land and ocean, which go beyond but are compatible with Western scientific approaches and can help us better manage the ocean (and land) (McAllister et al., 2023; Hikura, 2020); Uncle Sol (Solomon Pili Kahoʻohalahala) whose powerful voice at the International Seabed Authority (ISA) and in other fora urges leaders and participants to focus on culture and to consider the potential impacts of seabed mining on cultural connections and heritage (see Chap. 4, this vol.; Segalov, 2023; Pili Kahoʻohalahala, 2024); and scholar 'Atu Emberson-Bain with her sons (Siale and Menino Bain-Vete), daughter (Anga'aefonu (Fonu) Bain-Vete) and grandchildren (Tahi and Mahina Bain-Vete) who received the 2024 UK Rubery Book of the Year Award for their children's and

[1] The term Oceania refers to all the Pacific Islands and includes Aotearoa/New Zealand and Australia. Although the term was initially coined during colonial times it has been reclaimed (and popularised through the work of Epeli Hau'ofa) to emphasise the connections between Pacific peoples across the vast ocean, past and present, and to break away from narratives of smallness, isolation and vulnerability imposed by Western perspectives of the Pacific Islands.

[2] Different communities across Oceania have traditions of calling sea creatures using a range of techniques and rituals. The calling can be done for fishing or other purposes.

young adults' book *Cries from the Moana* (2023), about the dangers of deep sea mining, and provide a testament to the tradition and potency of intergenerational storytelling including in print (Fiji One News Team, 2024). These voices echo those of coastal and ocean community members whose lives are centred around the sea, some of whom could experience the direct threats of DSM.

As noted by Julian Arato et al. (2024) and Uncle Sol,[3] culture has not featured in the global discussions about deep sea mining, even though it is a central concern of communities in different parts of the world (DOSI, 2021). Connections to the ocean may be historical, spiritual, ancestral, practical and/or use based, but they are generally a combination of all, especially for communities in Oceania. Arato and colleagues raise the issue of the distinction between the use-value and the identity (sacred, spiritual, ancestral, genealogical) value of the ocean held by communities as posing both challenges and opportunities in the context of deep sea mining and the BBNJ Agreement (Arato et al., 2024). A challenge they suggest is that the use value might take away from the identity value by permitting room for negotiation rather than 'allow[ing] for a wholesale rejection of competing claims… and … prevent[ing] most other types of engagement with the deep seabed' (p.69) at least in the context of international law.

Although, as stated, the authors are referring to this distinction from an international law viewpoint in a discussion on Indigenous approaches in areas beyond national jurisdictions, it may also be a useful prism through which to consider the current opposing views about DSM in the Pacific. The views split between those fighting against it from a cultural values and livelihood perspective (which are intimately linked) and those, particularly the pro-DSM governments of Papua New Guinea, Cook Islands, Nauru, Kiribati, and Tonga (along with their associated mining company), who are convinced of the economic benefits of DSM. Their focus on use-values of the ocean tend to negate or, at best, ignore the identity values put forward by communities. This is not altogether surprising as use-values generally permeate government ocean policy in the Pacific in spite of statements about culture.[4] In the case of DSM, this is amplified by Pacific states' legal ownership of seabed mineral resources within Economic Exclusive Zones (EEZ)—a form of ownership which effectively denies communities from any benefit sharing, while leaving them with the burden of potential impact.

[3] Uncle Sol states, 'When I challenged The Metals Company on the absence of cultural considerations in their deep sea mining proposal presentation, they answered that they hadn't considered it prior to me coming to the ISA. They say they now plan to address this issue, but still do not know how to do so'. See: Pili Kahoʻohalahala (2024).

[4] Vanuatu is the only Pacific country which has placed culture at the foundation of its ocean policy. It is also strongly opposed to DSM.

8.2 EEZs of the Pacific

8.2.1 Papua New Guinea

Conflict between pro-DSM governments and communities, particularly in the case of prospecting[5] within national EEZs, has been an ongoing feature of DSM in Oceania, where communities view culture and environment as one and central to their lives for both identity and (sustainable) livelihood.[6] This is exemplified by the case of Papua New Guinea with the Solwara 1 prospecting's first phase in 2011. Solwara 1 was Oceania's first attempted foray into DSM, with prospecting taking place in the Bismarck Sea between the islands of New Ireland and New Britain. This first attempt failed, as Nautilus Minerals, the Canadian mining company which had partnered with the PNG government, declared bankruptcy in 2019. Opposing communities and civil society established the Alliance of Solwara Warriors in 2016, a group which has aimed to put a complete stop to DSM in PNG and the Pacific citing cultural and livelihood values and practices. Prospecting is now on the rebound in the Bismarck Sea with new activity underway via Deep Sea Mining Finance (DSMF), the company which took over Nautilus Minerals in 2019. The PNG government has seemingly approved this new prospecting effort in a bid to reverse the financial losses (around USD125 million) incurred through its partnership with Nautilus Minerals.[7] The venture is once again threatening the cultural wellbeing of neighbouring communities as illustrated by a PBS NewsHour broadcast series of interviews. In one of these, Chris Malgan, a traditional shark caller and chief of Kono village in New Ireland (located near where the prospecting is taking place), states, 'The ocean is our life. For us, the coastal people, it is our garden. We catch fish and share with others in the village without being paid. When we catch fish and sharks, we share with other families who need it to spread love and kindness' (Marx & Kiernan, 2024b). What Malgan and other community members fear is the destruction of their entire way of life, which is already being impacted by logging and the coastal impacts of climate change. DSM is the nail in the coffin: 'If this project goes ahead, the coastal communities won't be able to catch fish or shark anymore... and we will lose our culture that has been maintained for centuries' (Kessler, 2024).

'In our culture, a shark is seen as a blessing': the connection with sharks and shark calling discussed above is not unique to New Ireland's coastal communities, as shown through the recently published story of Uein Buranibwe and Temaei

[5] Prospecting in deep sea mining refers to the search for minerals which includes the use of machinery and equipment creating disturbance at sea.

[6] As Uncle Sol states, 'In our culture, there is no separation between us and nature—it is who we are. No matter where we are in the deepest part of the ocean, the surface of the waters, to the land and the mountains and the sky; we are part of all of it. So when industries are plotting to destroy the area of creation and have no regard for that part of our identity, it hurts'. See Pili Kahoʻohalahala (2024).

[7] A PBS NewsHour segment, the third and last in the series, raises doubts about the approval process within the PNG government. See: Marx and Kiernan (2024a).

Tontaake (Boucher et al., 2024). Stranded at sea for 33 days, the fishermen from Kiribati lassoed a shark to guide their small boat to Namdrik Atoll in the Marshall Islands. It is through the memory of his father's stories about using sharks to 'pull his canoe back to shore' when returning home from fishing, that Uein recalled that 'sharks like to swim towards land'. Upon reaching Namdrik, the fishermen discovered a genealogical link—Temaei's uncle, lost at sea decades earlier, had settled there, forming a new family branch. These connections, both physical and spiritual, reveal the unifying role of sharks and other marine species/ancestors as they connect Oceania's people, cultures, narratives and histories, tying together navigation, survival, and kinship across the Pacific.

8.2.2 Ecology and Culture

The cultural connection with sharks is intertwined with their ecological and geological significance. Shark teeth, often found on seamounts, can serve as nuclei for polymetallic nodules and cobalt-rich crusts, linking these iconic creatures to the frontier of deep sea mining. For Māori, the *pounamu* or greenstone of Aotearoa/New Zealand originate as soft matter inside a shark, hardening upon exposure to air (Nichols, 1992).[8] The geological connection between shark teeth, whale bones, crustacean shells, and ore formation reveals an intricate interplay of marine biodiversity, oceanic processes, and cultural traditions—an interplay now threatened by DSM. In Oceania, sharks are not just keystone species but cultural icons, embodying both heritage and ecological vitality. Acting as navigators, ecosystem indicators, and geological connectors, their presence on seamounts signifies the health of these marine habitats. This triple role of Ocean Heritage—ecological, geological and cultural—amplifies the urgency of protecting seamounts and the species they sustain.

In her rich discussion of the 'sea of voices' of stakeholders in PNG during the initial Solwara 1 experiment with Nautilus, Haley Phillips (2019) highlights the discordance between the economic use perspective put forward by the government and Nautilus, and the community position which draws on an 'ocean-based lifeworld'.[9] She demonstrates how the cultural relationship of communities with the ocean is negated by both Nautilus and the government through a range of

[8] This report chapter on sharks examines the cultural and ecological significance of sharks across the South Pacific, highlighting their roles in indigenous folklore and as keystone species in marine ecosystems. Nichols discusses traditional beliefs, such as the worship of sharks as ancestral spirits in Melanesian islands like the Solomon Islands and Papua New Guinea, and the Fijian legend of Dakuwaqa, a mythical shark deity. The report underscores the importance of integrating cultural perspectives into conservation efforts to ensure the sustainable management of shark populations in the region.

[9] Phillips draws on Ingersoll's (2016) notion of 'seascape epistemology', which is 'an approach to knowing presumed on a knowledge of the sea, which tells one how to move through it, how to approach life and knowing through the movements of the world. It is an approach to knowing through a visual, spiritual, intellectual, and embodied literacy of land and sea' (Ingersoll, 2016, pp. 5–6).

tactics, including a dismissive Environmental Impact Statement and shoddy consultation process, statements to the effect that local people are not connected to the sea beyond the shore, and reaffirmations of state power over the ocean seabed. She posits that 'like Nautilus, the national government has much to gain by refusing to acknowledge any cultural connection its citizens have to the Solwara1 site' (Phillips, 2019, p. 13) as it seeks to legitimise its own economic interests and control of seabed resources. This phenomenon is also described by John Childs in his assessment of Solwara 1 (Childs, 2019). He notably illustrates how the 'distancing of the deep ocean' from people and culture by the proponents of DSM, clashes with the cosmological world of coastal communities. Beyond the environmental impacts, Childs describes how DSM risks disturbing the spirits, the 'masalai—some of which are understood as guardians of the seabed and its resources' (Childs, 2019). Ignoring this factor, he argues, may not only marginalise the communities but also lead to 'spiritual turmoil'. He further notes how the removal of minerals, which, from the perspective of local communities, have been created by spiritual beings, is problematic: 'Thus, as copper "resurfaces" in the objects and technologies of the future—in batteries and wiring—it also carries a spirituality from the region where it originated'. The arguments put forward by governments and mining companies that DSM will have fewer impacts on communities than land-based mining, seem, in light of the world views of these communities, misguided and disingenuous. Acknowledging these cultural relationships would lead to a more balanced approach to DSM but the willingness to do this is currently absent.[10]

The profound spiritual turmoil caused by resource extraction is powerfully articulated by Katerina Martina Teaiwa in *Consuming Ocean Island* (2014). Teaiwa illustrates how the spirit of Banaba was extracted alongside its phosphate, fuelling global agricultural production while leaving the Banabans displaced and spiritually unmoored.[11] This echoes the experiences of coastal communities affected by DSM, where the disruption of *masalai*—spiritual guardians of the seabed—creates a similar cosmological imbalance. In both cases, the commodification of sacred elements for global gain undermines the intricate relationships between people and their environments. For the Banabans, their land (*te aba*) was not merely a resource but a spiritual and cultural cornerstone; its loss resulted in profound grief and cultural dispossession. Similarly, in Papua New Guinea, the removal of seabed minerals is seen as extracting not just physical wealth but spiritual essence, alienating communities from their cosmological worldviews. Ignoring these spiritual dimensions

[10] In the case of Papua New Guinea, the lack of transparency around the current Solwara 1 phase, as documented by PBS NewsHour does not augur well for communities. See: Marx and Kiernan (2024a).

[11] This work explores the cultural, environmental, and spiritual impacts of phosphate mining on Banaba, highlighting how the extraction of phosphate not only fuelled global agricultural production but also displaced the Banaban people and severed their connection to their homeland. Teaiwa critiques the commodification of sacred land and emphasizes the ongoing consequences of colonial resource extraction for Indigenous communities.

perpetuates marginalisation and cultural erasure, highlighting the urgent need for governance models that honour and incorporate Indigenous cosmologies.

8.2.3 Food, Culture, and the Environment

The intimate relationship between people, food, and the environment further illustrates the harm of extraction. On Banaba and Nauru, phosphate mining fuelled agricultural growth in nations like Australia, New Zealand, and Japan, but at a devastating cost. While the fertile soils of distant lands prospered, Banabans were left with barren landscapes and displaced communities. The voices of Banabans were silenced as British, Australian, and New Zealand investors prioritised profits, embodying the deep inequities of extractive industries where distant gains are built on local suffering.

In Fiji, the wisdom passed down through generations about fish poisoning (*kana ni ika dauveve*) offers a powerful example of how Pacific communities have long navigated environmental challenges. Fijian Traditional Ecological Knowledge (TEK) embodies a deep connection to the sea and its rhythms, guiding communities in protecting their food systems and avoiding harm. This knowledge includes identifying toxic fish species, understanding seasonal patterns tied to the spawning of *balolo* (edible seaworms), and recognising specific reefs as danger zones. For instance, reefs in Kadavu and Lau, marked by abundant *bulewa* (soft coral), are known as hotspots for toxic fish. Folk tests and traditional remedies, preserved through storytelling and practice, demonstrate the resourcefulness and adaptability of Fijians in responding to these risks. By blending wisdom with resilience, these practices sustain a harmonious balance with the ocean, reflecting a relationship that values respect and coexistence over exploitation.

This profound connection to the marine environment highlights the critical role of cultural knowledge in safeguarding food systems and living in harmony with nature. As the Pacific faces mounting threats like DSM and climate change, the lessons embedded in TEK offer a vital roadmap for resilience. As emphasised in Lako et al., 2023, fish poisoning remains a growing concern in Fiji, exacerbated by rising sea temperatures, cyclone seasons, and environmental disruptions.[12] Heat-resistant toxins such as ciguatoxins, palytoxins, and tetrodotoxins pose serious risks to food safety and public health, challenging conventional mitigation methods. TEK provides essential strategies to navigate these challenges, including identifying toxic hotspots like the Senimuna reef in Kadavu and Kabara reef in Lau, recognising seasonal toxicity patterns during balolo spawning, and employing folk tests to detect danger.

[12] The study underscores how disruptions in marine environments impact food safety and public health. It also emphasises the importance of blending traditional knowledge with scientific approaches to safeguard food systems in the face of environmental challenges.

Interestingly, Lako and colleagues also highlight the role of land-based activities in exacerbating marine toxicity. Copper mining near Udu Point in Vanua Levu and mahogany plantations on Cicia Island in the Lau Group are linked to a higher incidence of fish poisoning in those areas. Interviewees recounted a tragic incident in the 1990s where two individuals died after consuming turtle meat from a village near the mining site, suggesting a potential connection between mining activities and the contamination of marine life. Many Fijians believe that disruptions to the intricate relationships between the environment, land, and people can trigger fish poisoning, reflecting the holistic perspective embedded in TEK.

This worldview underscores the interconnectedness of human actions and natural systems, reminding us of the need to honour and integrate such knowledge to foster ecological balance and community well-being. These adaptive practices illustrate the enduring relationship between Fijians and their marine environment, highlighting the strength of cultural knowledge in addressing modern environmental challenges. By valuing and integrating TEK, Pacific communities can preserve their food systems, maintain ecological balance, and sustain their cultural identity in the face of an uncertain future.

This resilience is not new; historical parallels can be drawn to Polynesian migration, where environmental factors like ciguatera fish poisoning likely compelled communities to make critical survival decisions, as explored in Teina Rongo's research (Rongo et al., 2009).[13] Just as Polynesians relied on navigation and a deep understanding of their environment to escape regions of toxicity and relocate to areas of abundance, Fijians today draw upon TEK to navigate modern challenges. Preventive measures, such as folk tests and the removal of toxic organs, alongside herbal treatments, reflect the same ingenuity and adaptability that have long safeguarded Pacific food systems in the face of ecological disruptions.

Western science highlights the critical risks associated with DSM. Drazen et al. (2020) underscore the potential harm posed by sediment plumes and noise to midwater organisms and essential ecological processes, such as carbon sequestration and nutrient cycling. These processes are vital for maintaining global and regional environmental health, yet their potential disruption by DSM remains underexplored in dialogues with Indigenous Peoples and Local Communities (IPLCs). Bridging this gap is crucial, as DSM directly threatens food systems and ecological stability in the Pacific. Sediment plumes not only disrupt marine habitats and alter food webs but also endanger migratory species that are integral to Pacific Islanders' diets, livelihoods, and cultural identities.

These threats echo the broader vulnerabilities seen in historical cases such as Banaba and Nauru, where resource extraction led to environmental degradation, displacement, and cultural loss. Even Polynesian migration—driven in part by ecological crises—offers lessons about resilience and adaptation in the face of environmental challenges. These historical and ongoing examples highlight the urgency of

[13] See also how the article includes a detailed map of Polynesian migration routes, offering valuable insights into navigation, food security, and the historical connectivity of oceanic communities near regions like the Clarion-Clipperton Zone.

integrating traditional ecological knowledge, respecting Indigenous rights, and adopting precautionary governance frameworks. Such approaches are essential to balancing sustainable development with the preservation of cultural and ecological integrity, ensuring that IPLCs remain active stewards of their oceanic 'sea of islands'.

8.2.4 Cook Islands

DSM in national waters of the Cook Islands is also a profoundly divisive issue. In 2022, the government issued exploration licenses to three mining companies, one of which it co-owns and the Cook Islands Seabed Minerals Authority has established DSM regulations (Cook Islands News, 2023). As the 52nd Underwater Mineral Resources (UMR) Conference, 'the world's largest gathering of ocean mineral stakeholders', was about to take place in Rarotonga in September 2024, a poll conducted by Te Puna Vai Marama (the Cook Islands Centre for Research), found that '85% of respondents agreed that the deep ocean held cultural and spiritual significance for Cook Islanders' and that a majority of respondents were opposed to DSM exploration with 64% opposed to the mining itself. Among all respondents, 64% were against DSM on the basis that the 'deep ocean has cultural and spiritual significance for Cook Islanders' (Tuari'i & Worth, 2024, p. 9).

This poll, conducted online in both Cook Islands Māori and English, marked the first time the public was directly asked their opinion on DSM. Many opposing DSM feel that there has not been a robust consultation about mining in the country and that the mining sector has infiltrated Cook Island's society and economy. An example of this is the funding of many activities including, in what could be seen as a cynical move, the annual Cook Islands cultural festival, the Te Maeva Nui, and other cultural initiatives including an immersion Cook Islands Māori pre-school (Fuatai, 2024). As Louisa Castledine, who hails from Manihiki, which is one of the areas in Cook Islands with the richest mineral deposits, states '… the industry has secured the support of some of our foremost cultural and spiritual leaders by funding their organisations or initiatives' (Fuatai, 2024). In response to this co-opting of culture, a group of individuals, CSOs (including conservation organisations such as Te Ipukarea and Kōrero O Te 'Ōrau, and Te Manava Ora cultural group) and businesses, and with Casteldine and prominent cultural and environmental figures such as Dr Teina Rongo, formed Ocean Ancestors and staged the At Sea, Sail Out Rally (Protect our Ocean: Think Deeply) protest in Rarotonga during the UMR conference. The rally took place on the water with the traditional double hulled canoe, the *Vaka Marumaru Atua*, paddling canoes and other boats as well as on land. Whales, including a mother and her calf, came to the ocean area and right into the harbour including alongside a mining exploration vessel when the Ocean Ancestors were preparing for their rally. The organisers

consider this a strong sign of support and mutual recognition: 'that's our ancestors visiting us'.[14]

The rally reflects the deep-rooted connection between the people and their marine environment, a relationship that goes beyond material resources to encompass cultural identity, spiritual values, and traditional practices. The Manihiki Plateau, an oceanic plateau, exemplifies this profound relationship. Formed through volcanic activity more than 120 million years ago, its unique geological features, including seamounts, ridges, and knolls, support fragile ecosystems with sediment-eating organisms, sponges, and other marine species. Its proposed designation as an Ecologically or Biologically Significant Marine Area (EBSA) underscores its importance as a biodiverse yet vulnerable environment.

The plateau's ecological richness stands in stark contrast to the risks posed by deep sea mining, which targets its polymetallic nodules rich in manganese, nickel, cobalt, and other metals. Mining activities threaten to disrupt the seafloor, release harmful pollutants, and destroy habitats that remain poorly understood, with consequences that could reverberate throughout the marine food web. These risks have sparked concern among Cook Islanders, including residents of Manihiki, who argue that mining outside undermines local values. For the Manihiki community, the ocean is a source of spiritual connection, cultural identity, and livelihood, and protecting it is integral to ensuring the well-being of future generations.

Castledine, the spokesperson for Ocean Ancestors, argues that DSM serves external interests and is driven by overconsumption and capitalism. She emphasises that the Cook Islands are already facing the climate crisis, and that mining would further jeopardise their environment and way of life. The Cook Islands have a strong tradition of sustainable resource management, exemplified by practices like *ra'ui* and *tāporoporo*. Castledine believes that the extractive nature of deep sea mining would disrupt these traditional practices and negatively impact the natural resources upon which Cook Islanders depend. Castledine criticises the government's consultation process, arguing that it has been inadequate and lacks clarity regarding the potential impacts of deep sea mining (Fuatai, 2024). She recounts how her own family, despite attending government meetings, still has limited understanding of the environmental and social consequences. This lack of information is particularly concerning given the presence of a significant polymetallic nodule deposit near Manihiki, making it a prime target for mining companies. Castledine points to the tactics employed by the DSM industry, which she describes as a form of neo-colonialism. The strong emphasis on family relationships and collective cultural values, she explains, is being manipulated by companies who are seeking to gain a foothold in the islands. She argues that the mining industry is leveraging the Cook Islands' social structure to their advantage. Castledine highlights the use of euphemisms like 'harvesting minerals' and 'seabed minerals development' by the government and mining advocates to

[14] Pers. comm. June Hosking, Executive President, Te Ipukarea Society (03/12/2024).

downplay the negative connotations associated with mining. She believes this is a deliberate attempt to obscure the potential environmental damage and appease concerns.[15]

8.3 The High Seas

Transitioning from the focus on Exclusive Economic Zones to the high seas highlights the interconnected nature of Oceania and its cultural and ecological significance. The Clarion-Clipperton Zone (CCZ), situated beyond national jurisdictions, exemplifies the high seas as a shared space of immense opportunity and vulnerability. Unlike DSM activities within EEZs, which often centre on state-driven economic benefits, the high seas necessitate a broader, global approach—one that integrates ecological preservation, equitable resource sharing, and a recognition of the cultural and spiritual connections that Pacific communities maintain with the ocean. This perspective underscores the need for collaborative governance frameworks that honour the interconnectedness of Oceania while safeguarding its future.

The Clarion-Clipperton Zone, located between Hawaii and Mexico and bounded by the Clarion and Clipperton Fracture Zones, epitomises the tension between global economic ambitions and ecological and cultural preservation (Sharma, 2017). Since its discovery in 1954, the CCZ has become a focal point for DSM, with nations like Nauru, Tonga, and Kiribati partnering with contractors such as Nauru Ocean Resources Inc. (NORI), Tonga Offshore Mining Limited (TOML), and Marawa Research and Exploration Ltd. These companies are all subsidiaries of a global north seabed mining company, The Metals Company. Other exploration contracts in the CCZ involve countries like Belgium, China, France, Germany, Japan, Korea, the United Kingdom, and Singapore, alongside the InterOceanMetal Joint Organization, representing Bulgaria, Cuba, Czech Republic, Poland, Russia, and Slovakia. These activities are overseen by the International Seabed Authority (ISA), tasked with balancing the competing demands of economic development, environmental protection, and the cultural and spiritual values tied to the ocean (see Chap. 4, this vol.).

Through the lens of Epeli Hauʻofa's philosophy of the 'Sea of Islands' (1994), Oceania emerges not as fragmented territories but as a vast, interconnected region unified by its peoples, cultures, and histories. Hauʻofa's vision challenges the colonial legacy of division and reductionism, emphasising instead the fluidity and

[15] See also, Hosking (2024). This letter critiques the lack of transparency and community participation in seabed mining decision-making processes in the Cook Islands, highlighting the concentration of power in the hands of a few and questioning the Seabed Minerals Authority's claim of a 'social licence' to mine. Concerns include insufficient consultation with traditional leadership, the Church, and the public, environmental risks to biodiversity, and the prioritisation of mining legislation over other environmental issues. The letter underscores the need for genuine consultation and transparency.

connectivity of the Pacific. In this context, the high seas, including areas like the CCZ, are not isolated expanses but integral parts of the shared heritage and identity of Pacific Island communities. They are spaces where cultural narratives, traditional knowledge, and ecological stewardship converge, offering a counterpoint to extractive models that prioritize short-term profit over long-term sustainability.

As DSM extends its reach from EEZs to the high seas, it threatens to sever the intricate cultural and ecological relationships that Pacific peoples have nurtured for generations. The 'Sea of Islands' philosophy challenges reductionist views of the Pacific, portraying it not as fragmented territories ripe for exploitation but as an interconnected, vibrant expanse central to Oceania's identity, livelihoods, and spiritual well-being. This perspective underscores the ocean's role as a shared heritage that transcends jurisdictional boundaries and demands a holistic approach to stewardship.

The push for mining in the CCZ reflects an enduring colonial mindset that prioritises economic gain over ecological and cultural integrity. Mining threatens to replicate the extractive practices that have left lasting scars on Pacific communities. For example, phosphate mining on Banaba Island devastated 90% of its surface, transforming it into an industrial wasteland and forcibly displacing the Indigenous Banaban people. This relocation fractured their cultural ties to their ancestral homeland, creating intergenerational trauma that continues to impact their identity and well-being. In the book *Inflamed*, Rupa Marya and Raj Patel (2022) argue that colonial extractivism is not merely a relic of the past but an ongoing systemic force that disrupts ecosystems, displaces communities, and exacerbates inequality and health disparities. Without careful governance and cultural sensitivity, the CCZ risks becoming another chapter in this exploitative legacy, reinforcing patterns of ecological destruction and cultural erasure in the pursuit of short-term economic benefit.

Beyond its mineral wealth, the CCZ is one of Earth's most biodiverse regions, with an astounding 70–90% of collected species new to science. The polymetallic nodules that attract mining interest are far more than inert resources; they serve as critical habitats for diverse life forms, ranging from microbial communities to deep-sea invertebrates. This biodiversity is deeply interwoven with the cultural and spiritual worldviews of Pacific communities, who see the ocean as a living, dynamic entity sustaining both ecological and human well-being. Without deliberate governance measures, the CCZ risks becoming another chapter in the legacy of extractive exploitation, undermining both ecological resilience and cultural heritage.

Tongan scholar Tēvita O Ka'ili's discussion of *vä*, or 'sacred relationality', provides a profound lens for understanding the interconnectedness of people and their environment (Ka'ili, 2005, 2017).[16] *Vä* underscores the inseparability of cultural

[16] Tēvita O Ka'ili has extensively explored the concept of vā, or 'sacred relationality', in his scholarly work. In his 2005 article, Ka'ili examines how Tongans maintain and nurture social relationships and spaces, emphasizing the importance of vā in Tongan culture. Additionally, in his 2017 book, *Marking Indigeneity: The Tongan Art of Sociospatial Relations,* Ka'ili delves into the Tongan art of sociospatial relations, further elucidating the significance of vā in the context of Tongan identity and community.

identity and ecological stewardship, offering a framework that highlights the need for governance models that respect the cultural and ecological dimensions of the high seas. This relational perspective calls for global policies that balance innovation with the preservation of cultural and ecological legacies, ensuring that the ocean remains a shared space of connectivity and life for future generations. Similarly, the Māori principle of *whakapapa* links all things through ancestral lineage, establishing responsibilities to both people and the natural world. From these relationships arises *kaitiakitanga*—a duty to protect and manage ancestral domains for future generations.

Oceania's leadership in community-based protected areas exemplifies the integration of these values into modern conservation initiatives. These efforts prioritise sustainable use and conservation, drawing on cultural kinship, customary governance, and ecological stewardship to achieve a balance between development and preservation. In Aotearoa, *kaitiakitanga* embodies this balance, intricately connected to *rangatiratanga*, the right to self-determination outlined in Te Tiriti o Waitangi (Jackson et al., 2017). Together, these Indigenous frameworks offer powerful models for high seas governance, demonstrating that economic innovation can and must coexist with the preservation of cultural and ecological legacies, intertwined with the cultural and spiritual perspectives of many Pacific communities, who view the ocean as more than a resource—it is a living, dynamic entity that sustains both ecological and human well-being (McAllister et al., 2023).

These risks compel us to adopt a broader perspective—one that aligns scientific research with cultural knowledge and wisdom. The CCZ is not just a repository of minerals; it is part of a larger, interconnected oceanic system imbued with cultural narratives and ecological significance. Pacific philosophies remind us that the ocean is a shared space of connection and responsibility. As stated, Pacific Islanders have long mapped sea routes, named places across vast distances, and cultivated spiritual relationships with the ocean. Far from being resource-poor, Oceania's wealth lies in its ecological and cultural knowledge systems, which reflect a profound understanding of the natural world and its interconnectedness with human life. Integrating these cultural perspectives with precautionary scientific approaches offers a pathway to protect the CCZ's fragile ecosystems while honouring the deep relationship between Pacific communities and their ocean. In this way, we can envision a future where both biodiversity and cultural integrity are safeguarded for generations to come.

Colonial and industrial narratives, however, have overshadowed these perspectives. The CCZ, like much of Oceania, has been framed as a resource-rich frontier ripe for exploitation, perpetuating the history of environmental extraction that devastated Indigenous ways of life. For instance, the loss of Indigenous languages—resulting from colonial assimilation policies—has severed critical connections between people and their environments. These languages often encoded sophisticated ecological knowledge, contributing to the erosion of cultural identity and well-being.

Resistance to DSM is deeply rooted in Oceanian cultural identity and place-attachment. The Kumulipo, a Hawaiian creation chant, exemplifies this relationship. It recounts the origins of the world, emphasising the interconnectedness of humans and the ocean. Uncle Sol uses the Kumulipo to advocate for the protection of the deep sea, asserting that its genealogical ties to Hawaiians make it a sacred site, not a vacant expanse for exploitation. This perspective aligns with Indigenous knowledge systems that emphasise holistic approaches to health and well-being. *Inflamed* contrasts this with Western medicine's reductionist approach, which isolates the individual body from its ecological and social context. Indigenous frameworks, in contrast, reveal the intricate web of relationships connecting human health with environmental health.

8.4 Conclusion

The transformation of marine ecosystems through industrial activities like mining and shipping poses huge risks in the Pacific region.

The disruption of environmental and food connectivity, as seen in the links between land-based pollution and ciguatera fish poisoning (CFP) in Fiji and the Cook Islands, resonates with broader challenges posed by sediment plumes generated by DSM operations. Just as terrestrial contaminants from mining activities and agricultural runoff can destabilise marine ecosystems by promoting harmful algal blooms, DSM sediment plumes extend these risks to previously undisturbed deep-sea environments, upon which communities rely for their livelihoods and identity. The scale of these plumes—discharging tens of thousands of cubic meters of sediment daily, as in the Clarion-Clipperton Zone—threatens migratory species and regional food chains, critical to both ecological integrity and the wellbeing, identity and subsistence needs and practices of Pacific communities.

But what are some solutions and examples of management? Oceania leads the world in community-based protected areas, which account for a large portion of all protected sites in the region. These initiatives prioritise sustainable use alongside conservation goals, rooted in cultural kinship, customary governance and ecological stewardship. They offer a model for balancing development with the preservation of cultural and ecological integrity. Similarly, Vanuatu demonstrates how integrating traditional knowledge and values into national policies strengthens ocean stewardship. Vanuatu's Marine Spatial Plan (MSP) serves as a cornerstone of Vanuatu's inclusive approach, incorporating zoning systems that empower local communities through Community Conservation Areas and no-take zones, while protecting biodiversity and supporting sustainable resource use. Traditional knowledge, documented through initiatives like the Vanuatu National Traditional Knowledge Indicators Booklet, plays a pivotal role in guiding climate adaptation and resource management. By embedding Indigenous practices such as weather forecasting

based on environmental cues into contemporary frameworks like the MSP, Vanuatu demonstrates how Indigenous knowledge can enhance resilience and ecological stewardship.

Addressing the multitude of challenges requires listening to, respecting, and integrating cultural approaches and traditional ecological knowledge into ocean governance frameworks. To conclude, we invite readers to reflect on the words of the Pacific Theological College Principal, Rev. Professor Upolu Vaai and his call on all churches, all Christians, all Pacific Islanders to stand together to protect the ocean:

> We don't have the ocean, we are ocean. We don't have the land, we are land. Watery to our very core. Earthy to our very being. The Moana is symbolic of the Spirit. Dynamic. Connected. Alive. Yet Groaning. Constantly seeking our hand of compassion in the age of destruction. The moana groans for restoration. Yet the Spirit comes to rouse the people. The Spirit births new voices. She sparks movements of resistance. She creates currents of contradictions to exploitation. She disturbs our complacency. She inspires new moana consciousness. She makes us passionate people for the one groaning. With a wild and unruly practice of hope. Lord God, reconnect us to the moana. Show us a new Alaimoana. A new pathway. Guided by stars of hope that guided our forebears. Create in us a new consciousness of care and connectedness guided by your love. Reconnect us to our watery home. Our fluid identity. In the name of the God who liberates us from the gods of money. Amen! (Raj, n.d.)

References

Arato, J., Jaeckel, A., & Lixinski, L. (2024). On the cultural stakes of deep seabed mining. *AJIL Unbound, 118*, 67–71. https://doi.org/10.1017/aju.2024.9

Boucher, D. L., Macumber, B., & James-Proud, M. (2024, October 25). Kiribati fishermen tell how they survived being lost at sea for 33 days and solved a family mystery. *ABC Pacific*. https://www.abc.net.au/pacific/uein-buranibwe-was-lost-at-sea-for-33-days-story-of-survival/104491232.

Childs, J. (2019, February 19). Deep sea mining threatens indigenous culture in Papua New Guinea. *The Conversation*. http://theconversation.com/deep-sea-mining-threatens-indigenous-culture-in-papua-new-guinea-112012.

Cook Islands News. (2023, November 22). Sovereign rights not enough, regional cooperation needed for deep-sea mining: TIS. *Islands Business*. https://islandsbusiness.com/news-break/deep-sea-mining-3/.

DOSI. (2021). *The necessity of traditional knowledge for management of deep-seabed mining* [Policy Brief]. https://www.dosi-project.org/wp-content/uploads/072-DOSI-Policy-brief-Traditional-Knowledge-for-Management-of-DSM-long-V11.pdf.

Drazen, J. C., Smith, C. R., Gjerde, K. M., Haddock, S. H. D., Carter, G. S., Choy, C. A., Clark, M. R., Dutrieux, P., Goetze, E., Hauton, C., Hatta, M., Koslow, J. A., Leitner, A. B., Pacini, A., Perelman, J. N., Peacock, T., Sutton, T. T., Watling, L., & Yamamoto, H. (2020). Midwater ecosystems must be considered when evaluating environmental risks of deep-sea mining. *Proceedings of the National Academy of Sciences, 117*(30), 17455–17460. https://doi.org/10.1073/pnas.2011914117

Emberson-Bain, ʾA., Bain-Vete, M., Bain-Vete, S., & Bain-Vete, A. (2023). *Cries from the Moana*. Hakautapu Publishing.

Fiji One News Team. (2024, August 9). International acclaim for "Cries from Moana." *Fiji News*. https://fijionenews.com.fj/international-acclaim-for-cries-from-moana/.

Fuatai, T. (2024, November 23). Deep-sea mining's charm offensive in the Cook Islands. *E-Tangata*. https://e-tangata.co.nz/comment-and-analysis/deep-sea-minings-charm-offensive-in-the-cook-islands/.

Hau'ofa, E. (1994). Our Sea of Islands. *The Contemporary Pacific, 6*(1), 148–161. https://www.jstor.org/stable/23701593

Hikuroa, D. (2020). Matauranga Maori and its role in coastal management. In C. Hendtlass, S. Morgan, & D. Neale (Eds.), *Coastal systems and sea level rise: What we look for in the future*. New Zealand Coastal Society. https://www.coastalsociety.org.nz/assets/Publications/Special-Issues/SP4-Low-res-version.pdf

Hosking, J. (2024, October 11). Letter: Social licence for deep sea mining. *Cook Islands News*. https://www.cookislandsnews.com/internal/opinion/letters-to-the-editor/letter-social-licence-for-deep-sea-mining/.

Ingersoll, K. E. (2016). *Waves of knowing: A seascape epistemology*. Duke University Press.

Jackson, A. M., Mita, N., & Hakopa, H. (2017). *Hui-te-ana-nui: Understanding kaitiakitanga in our marine environment* (Sustainable Seas National Science Challenge Project). https://www.sustainableseaschallenge.co.nz/assets/dms/Reports/Hui-te-ana-nui-Understanding-kaitiakitanga-in-our-marine-environment/HTAN-report-2017.pdf.

Ka'ili, T. O. (2005). Tauhi vā: Nurturing Tongan Sociospatial Ties in Maui and Beyond. *The Contemporary Pacific, 17*(1), 83–114. https://www.jstor.org/stable/23721933

Ka′ili, T. O. (2017). *Marking indigeneity: The Tongan art of sociospatial relations*. *University of Arizona Press*. https://doi.org/10.2307/j.ctt1t89kr9

Ka'ili, T. (2024, February). *Fonua: Intersecting Cosmogony and Ecology* [2024 David O. Mckay Lecture]. https://speeches.byuh.edu/david-o-mckay-lectures/intersection-cosmogony-and-ecology.

Kessler, R.,2024. Opinion: Communities worry anew as PNG revives seabed mining plans. In Islands Business, March 21, 2024.

Lako, J. V., Naisilisili, S., Vuki, V. C., Kuridrani, N., & Agyei, D. (2023). Local and traditional ecological knowledge of fish poisoning in Fiji. *Toxins, 15*(3), 223. https://doi.org/10.3390/toxins15030223

Lala's LALINOA In-Depth Conversation with Tevita Ka'ili: Exploring Ancient Tongan Beliefs Before Christianity and Their Influence on Traditional Practices Today. (2024, September). [Broadcast]. Radio Nukuálofa 88.6FM. https://www.youtube.com/watch?v=Z-V3MfhRBRc.

Marx, W., & Kiernan, E. (2024a, September 19). Papua New Guinea leaders struggle to monitor deep-sea mining activities off its coast. [Broadcast]. In *News hour*. PBS. https://www.pbs.org/newshour/show/papua-new-guinea-leaders-struggle-to-monitor-deep-sea-mining-activities-off-its-coast.

Marx, W., & Kiernan, E. (2024b, September 18). Papua New Guinea locals concerned over deep-sea mining's impact on culture and environment. [Broadcast]. In *News hour*. PBS. https://www.pbs.org/newshour/show/papua-new-guinea-locals-concerned-over-deep-sea-minings-impact-on-culture-and-environment.

Marya, R., & Patel, R. (2022). *Inflamed: Deep medicine and the anatomy of injustice* (First paperback ed.). Picador.

McAllister, T., Hikuroa, D., & Macinnis-Ng, C. (2023). Connecting science to indigenous knowledge: Kaitiakitanga, conservation, and resource management. *New Zealand Journal of Ecology*. https://doi.org/10.20417/nzjecol.47.3521

Nichols, P. (1992). *Sharks* (No. 92/66). FFA (Pacific Islands Forum Fisheries Agency). https://spccfpstore1.blob.core.windows.net/digitallibrary-docs/files/69/690fca052ecccef8edf66bbf2e5fdfd6.pdf?sv=2015-12-11&sr=b&sig=e44UALp00Jao%2BDtPZo6uY7UmstP6mhT2dlNlSjyxkOY%3D&se=2025-06-05T20%3A40%3A58Z&sp=r&rscc=public%2C%20max-age%3D864000%2C%20max-stale%3D86400&rsct=application%2Fpdf&rscd=inline%3B%20filename%3D%22FFA_1992_066.pdf%22.

Philips, H. (2019). *A sea of voices: Deep sea mining and the Solwara 1 Project in Papua New Guinea* [MA Thesis, University of Waikato]. https://researchcommons.waikato.ac.nz/entities/publication/e60d590f-82d4-4e88-ab31-5f703e4d1be0.

Pili Kaho'ohalahala, S. (2024, December 16). *What global deep sea mining negotiations are missing*. Greenpeace International. https://www.greenpeace.org/international/story/66154/uncle-sol-pacific-voices-say-stop-deep-sea-mining/.

Raj, B. (n.d.). *A CALL FOR THE OCEAN – Pacific Theological College*. Retrieved December 16, 2024, from https://ptc.ac.fj/a-call-for-the-ocean/.

Rongo, T., Bush, M., & Van Woesik, R. (2009). Did ciguatera prompt the late Holocene Polynesian voyages of discovery? *Journal of Biogeography, 36*(8), 1423–1432. https://doi.org/10.1111/j.1365-2699.2009.02139.x

Segalov, M. (2023, May 21). Is it too late to halt deep-sea mining? Meet the activists trying to save the seabed. *The Observer*. https://www.theguardian.com/environment/2023/may/21/is-it-too-late-to-halt-deep-sea-mining-the-activists-trying-to-save-the-seabed.

Sharma, R. (Ed.). (2017). *Deep-sea mining: Resource potential, technical and environmental considerations*. Springer International. https://doi.org/10.1007/978-3-319-52557-0

Teaiwa, K. M. (2014). *Consuming Ocean Island: Stories of people and phosphate from Banaba*. Indiana University Press.

Torrente, F., Bambridge, T., Planes, S., Guiart, J., & Clua, E. G. (2018). Sea swallowers and land devourers: Can shark lore facilitate conservation? *Human Ecology, 46*(5), 717–726. https://doi.org/10.1007/s10745-018-0018-7

Tuari'I, M., & Worth, H. (2024). *Opinion poll on deep-seabed mining in the cook islands, Analysis Report 2024*.

Open Access This chapter is licensed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits any noncommercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if you modified the licensed material. You do not have permission under this license to share adapted material derived from this chapter or parts of it.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.